

Pharma Agentic AI – Analysis Report

Query: Identify scientifically justified repurposing opportunities for existing asthma and COPD molecules based on mechanism of action, inflammatory pathways (Type-2 high vs Type-2 low), and biomarker-driven patient subgroups in India.

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Report ID: 86477e98-a9f9-4f09-b654-9a9720fd0970

Executive Summary

No IQVIA data found for given filters. | No clinical trials found for given filters. | Repurposing opportunities for existing asthma and COPD molecules focus on addressing unmet needs in India, such as steroid-free options, safer formulations for the elderly, alternative therapies for steroid intolerant patients, low-dose options for pediatric asthma cases, and addressing adherence issues through once-daily inhalers. | Recent CHEST 2024 guidelines support early COPD screening for smokers through spirometry, citing evidence that underdiagnosis persists globally. Rural India highlights spirometry shortages, while patients opt for affordable DPI devices. Oral steroid usage remains suboptimal.

IQVIA Insights Agent

Summary: No IQVIA data found for given filters.

Details:

Table:

Clinical Trials Agent

Summary: No clinical trials found for given filters.

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Table:

Internal Knowledge Agent

Summary: Repurposing opportunities for existing asthma and COPD molecules focus on addressing unmet needs in India, such as steroid-free options, safer formulations for the elderly, alternative therapies for steroid intolerant patients, low-dose options for pediatric asthma cases, and addressing adherence issues through once-daily inhalers.

Details:

Documents:

- asthma_strategy_2024.pdf
 - respiratory_field_insights.pdf
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Highlights:

- **doc_name:**
C:\Users\vishw\OneDrive\Desktop\Pharma\backend\app\data\internal_docs\asthma_strategy_2024.pdf
page: 0 **snippet:** Asthma Market & Unmet Needs – Internal Strategy 2024 • High burden in India with increasing prevalence. • Despite many competitors, steroid-free options limited. • Elderly population ke liye safer formulations ki demand badh rahi hai. • Fixed-dose combinations ka uptake stable hai but expensive. • Opportunity: once-daily inhaler for better adherence.
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- **doc_name:**

C:\Users\vishw\OneDrive\Desktop\Pharma\backend\app\data\internal_docs\respiratory_field_insights.pdf

page: 0 **snippet:** respiratory_field_insights • Tier-2/3 cities me inhaler affordability ek major issue hai. • Steroid intolerance wale patients ke liye alternative therapies kam hain. • Many patients struggle with correct inhaler technique → low adherence. • Rural hospitals prefer cheaper DPIs over expensive MDIs. • Pediatric asthma cases me suitable low-dose options missing.

Key Opportunities:

- Repurpose long-acting bronchodilators (LABAs) for fixed-dose combinations with non-steroidal anti-inflammatory agents (NSAIDs) for steroid-free options.
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- Target elderly population with safer formulation of inhaled corticosteroids (ICS) or LABAs.
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- Develop once-daily inhalers for improved adherence in adult and pediatric asthma patients.
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- Repurpose phosphodiesterase-4 inhibitors (PDE4is) or janus kinase inhibitors (JAKis) for alternative therapies in steroid-intolerant patients.
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- Tailor low-dose inhalers for pediatric asthma cases, focusing on combination of ICS and LABAs.
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Key Risks:

- Potential liver enzyme induction with NSAIDs may increase liver function test (LFT) elevations and reduce LABA efficacy.
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- Concerns about increased cardiovascular risk with LABA-ICS combinations, particularly in elderly patients.
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- Evidence gaps in assessing efficacy and safety of PDE4is or JAKis in specific patient populations, including pediatric cases.
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- Potential for interactions between LABAs and other respiratory medications, affecting their efficacy and safety.
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Web Intelligence Agent

Summary: Recent CHEST 2024 guidelines support early COPD screening for smokers through spirometry, citing evidence that underdiagnosis persists globally. Rural India highlights spirometry shortages, while patients opt for affordable DPI devices. Oral steroid usage remains suboptimal.

Details:

Links:

- **title:** Respiratory guideline (synthetic) **url:** <https://example-guideline.com>
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- **title:** Respiratory RWE summary **url:** <https://example-rwe.com>
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